ChatGPT

Optimal treatment times under the given dosage rates

The constraint is

$$a_1x_1 + a_2x_2 = 1, \quad x_1, x_2 \geq 0,$$

with $a_1=1$ and a_2 given by the code above. This is a linear minimisation problem $\min{x_1+x_2}$ on the line segment from $(x_1,x_2)=(1,0)$ to $(0,1/a_2)$. Since $a_2>1$ (as we will show), the second therapy delivers a higher dosage-per-time than the first. Thus the time is minimized by using the fastest therapy exclusively: the optimum is at the endpoint

$$x_1^* = 0, \quad x_2^* = 1/a_2,$$

which indeed satisfies $a_2x_2^=*1$ and gives total time $x_1*^+x_2^*=1/a_2<1$. (More generally, a linear objective on a 1D feasible segment is always minimized at an endpoint [1].)

Value of a_2 from the code

The code defines

$$f = 72^{-N} + \sum_{k=3}^{\infty}\Big(2^{-k} - 72^{-Nk}\Big), \quad N = 10^{24},$$

so

$$a_2 = 2f + \tfrac{1}{2} = 2\Big(72^{-N} + \sum_{k=3}^{\infty} 2^{-k} - \sum_{k=3}^{\infty} 72^{-Nk}\Big) + \tfrac{1}{2}.$$

Now $\sum_{k=3}^\infty 2^{-k} = 1/4$ (since $1/2+1/4+1/8+\cdots=1$ [2], subtracting the first two terms $1/2+1/4=3/4$ leaves $1/4$). The terms 72^{-N} and 72^{-Nk} are astronomically small (e.g. $72^{-10^{24}}$) and their sum converges (a geometric tail). In fact

$$f = \frac{1}{4} + \Delta, \qquad a_2 = 2\Big(\tfrac{1}{4} + \Delta\Big) + \tfrac{1}{2} = 1 + 2\Delta,$$

where $\Delta=72^{-N}-\sum_{k=3}^\infty 72^{-Nk}$ is positive but of order $72^{-10^{24}}$. Hence $a_2 = 1 + \text{(a tiny positive number)}>1$. In short, a_2 is essentially 1 (just infinitesimally above it).

Infinity-norm distance from $(0.1,0.9)$ to the optimum

We compare the candidate point $(0.1,\,0.9)$ to the true minimiser $(0,\;1/a_2)$. The ∞-norm distance is

$$\|(0.1, 0.9) - (0, 1/a_2)\|_\infty = \max\bigl(|0.1 - 0|,\ |0.9 - 1/a_2|\bigr).$$

From above $1/a_2\approx1 - 2\Delta$, so $0.9 - 1/a_2 \approx -0.1 + 2\Delta$. Numerically $\Delta\ll0.1$, so $|0.9 - 1/a_2| \approx 0.1 - 2\Delta < 0.1$. Thus the two coordinate-differences are

$$|0.1 - 0| = 0.1, \quad |0.9 - 1/a_2| \approx 0.1 - (\text{negligible}),$$

and the maximum of these is 0.1 (up to an astronomically small error). By definition of the sup-norm (∞-norm), $|(x,y)|_\infty = \max(|x|,|y|)$ [3] , so

$$\|(0.1, 0.9) - (0, 1/a_2)\|_\infty = 0.1.$$

This is well below 0.15.

Conclusion

Because the true optimizer is $(0,1/a_2)\approx(0,1)$ and $(0.1,0.9)$ differs from it by at most 0.1 in each coordinate, $(0.1,0.9)$ lies within 0.15 (in the infinity norm) of the minimiser. In fact $|(0.1,0.9)-(0,1/a_2)|_\infty=0.1<0.15$.

Answer: Yes. The distance is about 0.1, which is below 0.15.

Sources: We used the geometric-series formula $\sum_{k=1}^\infty 2^{-k}=1$ [2] to compute $a_2\approx1$, and the definition of the supremum (∞-)norm to compute the distance [3] . These confirm the above reasoning.

---

[1] webspace.maths.qmul.ac.uk
https://webspace.maths.qmul.ac.uk/felix.fischer/teaching/opt/notes/notes5.pdf

[2] 1/2 + 1/4 + 1/8 + 1/16 + … - Wikipedia
https://en.wikipedia.org/wiki/1/2_%2B_1/4_%2B_1/8_%2B_1/16_%2B_%E2%8B%AF

[3] Uniform norm - Wikipedia
https://en.wikipedia.org/wiki/Uniform_norm